Mingyuan Zhang

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Education

Northeastern University *Sep. 2023 – Present*
Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu

- **Research Area:** Foundation Models, Model Efficiency, Multimodal Learning, Generative Models

Northeastern University *Sep. 2021 – Dec. 2022*
M.S. in Electrical/Computer Engineering

- **Concentration:** Computer Vision, Machine Learning and Algorithm

Xidian University *Sep. 2017 – Jun. 2021*
B.S. in Communication Engineering

Skills

Languages: Strong reading, writing and speaking competencies for English, Mandarin Chinese.

Coding: Python, MATLAB, C, Xcode.

Web Dev: HTML, CSS.

Misc.: Academic research, LaTeX typesetting and publishing.

Experience

Research Assistant *Boston, MA*
Northeastern University *Sep. 2023 - Present*

- **Large Language Models and Multimodal Learning**
  - Proposed Mask Fine-Tuning (MFT), a novel parameter-efficient adaptation paradigm that improves large language and vision-language models, consistently outperforming even full fine-tuning.
  - Developed a structured latent expert Mixture-of-Experts (MoE) framework for large language models to reduce inference FLOPs while maintaining or even improving model performance.
  - Conducted SoupLM, a learnable model integration framework that merges multiple large language and multimodal models into a unified model with improved generalization and negligible inference overhead.
  - Contributed to Cautious Next Token Prediction (CNTP), an entropy-aware decoding strategy that improves reasoning accuracy and reliability of large language models.
  - Collaborated on systematic studies on hallucination in video large language models, identifying root causes and advancing research toward more reliable multimodal systems.
- **Agents**
  - Contributed to an agent-centric framework for visual generation that leverages explicit structural guidance to bridge high-level intent and low-level synthesis.
- **Pruning and Distillation**
  - Proposed Sharpness-Aware Random Sparse Training (SA-RST) and Dual Lottery Ticket Hypothesis, enabling effective training of randomly initialized sparse neural networks without dense pre-training.
  - Leading a comprehensive study on model/data pruning and distillation aimed at improving model performance for foundation models.
- **Autonomous Driving and Robotics**
  - Investigated the integration of large language models into trajectory prediction, identifying key challenges and advancing language-driven motion understanding for autonomous systems.
  - Developing a backend-agnostic visual evidence agent for autonomous driving that actively selects minimal high-resolution regions to improve VLM decision quality under strict latency and token budgets.
- **Generative Models**
  - Developed LST-GAN, a spatial-temporal generative framework that synthesizes missing camera views to improve robustness and accuracy in 3D human pose estimation.

Research Intern *Remote*
Socure *Jun. 2025 – Aug. 2025*

- **Low-Level Vision**
  - Developed a computational framework for barcode reconstruction aimed at mitigating the effects of occlusion, blur, and partial data loss, thereby enhancing decoding robustness and accuracy.

Applied Scientist Intern *Remote*
AinnovationLabs *Dec. 2022 – Dec. 2024*

- **Healthcare AI and App Development**
  - Invented **1 patent** for sleep signal processing that identifies sleep stages and detects sleep disorders from complex acoustic signals.
  - Developed a mobile app for recording sleep patterns and consulting sleep-related questions, enabling large-scale multimodal data collection for sleep research.
- **Computer Vision and Human-Centered Perception**
  - Developed a spatio-temporal 3D human pose estimation model with a self-collected dataset for accurate single-person motion tracking.
  - Built a pedestrian re-identification system capable of recognizing the same individual across diverse environments.

Research Assistant *Nanjing, China*
Southeast University *Jun. 2019 - Aug. 2019*

- **Environmental Modeling**
  - Designed a prediction model for atmospheric environmental changes by integrating multimodal environmental data, improving forecasting accuracy and robustness.

Publications

Under Review / Preprint

[U-1] ***Mingyuan Zhang***, Yue Bai, Huan Wang, Yizhou Wang, Qihua Dong, Yitian Zhang, and Yun Fu, "Boosting Large Language Models with Mask Fine-Tuning". [Paper] [Code]

[U-2] ***Mingyuan Zhang***, Yue Bai, Yifan Wang, Yiyang Huang, and Yun Fu, "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models". [Paper]

[U-3] Yi Xu, Ruining Yang, Yitian Zhang, Yizhou Wang, Jianglin Lu, ***Mingyuan Zhang***, Lili Su, and Yun Fu, "Trajectory Prediction Meets Large Language Models: A Survey". [Paper] [Code]

[U-4] Yiyang Huang, Yitian Zhang, Yizhou Wang, ***Mingyuan Zhang***, Liang Shi, Huimin Zeng, and Yun Fu, "Distorted or Fabricated? A Survey on Hallucination in Video LLMs". [Paper] [Code]

[U-5] Yue Bai*, ***Mingyuan Zhang****, Huan Wang, Zhiqiang Tao, Kunpeng Li, and Yun Fu, "Random Sparse Networks Training with Sharpness-Aware Regularization".

[U-6] Yue Bai*, ***Mingyuan Zhang****, Handong Zhao, and Yun Fu, "Advancing Multi-Modal Foundation Models with Adapter Ensemble Strategies".

Conference

[C-1] Yizhou Wang, Lingzhi Zhang, Yue Bai, Mang Tik Chiu, Zhengmian Hu, ***Mingyuan Zhang***, Qihua Dong, Yu Yin, Sohrab Amirghodsi, and Yun Fu, "Cautious Next Token Prediction", ***ACL 2025***. [Paper] [Code]

Ongoing Work

[O-1] ***Mingyuan Zhang***, Yue Bai, Yi Xu, Chang Liu, and Yun Fu, "LST-GAN: Latent Spatio-Temporal Generative Adversarial Networks for Missing-View 3D Human Pose Estimation".

[O-2] Xu Ma, Jianglin Lu, ***Mingyuan Zhang***, and Yun Fu, "Agentic Visual Generation in the Late 2020s".

[O-3] Yue Bai*, ***Mingyuan Zhang****, Jiasen Lu, Zichen Zhang, and Yun Fu, "SoupLM: Model Integration in Large Language and Multi-Modal Models".